

5.2 Properties of the Matrix Inverse



Core skills you'll master in this section:

- ▶ Build geometric intuition: understand the effects of various inverse transformations intuitively from the perspectives of rotation, shearing, scaling, and more.
- ▶ Determine whether a matrix is invertible: a nonzero determinant is the necessary and sufficient condition for a square matrix to be invertible.
- ▶ The geometric meaning of a non-invertible square matrix: information is lost, as in operations that cannot be undone, such as dimension reduction or projection.
- ▶ Why a non-square matrix is not invertible: its input and output dimensions differ, so it cannot uniquely reflect the original information.
- ▶ The inverse of a chain of matrix products: the inverse of a product of invertible matrices is the product of their individual inverses in reverse order.
- ▶ When a diagonal square matrix is invertible, its inverse is the new diagonal matrix formed by taking the reciprocal of each diagonal entry.
- ▶ The inverse of a symmetric matrix is still symmetric; the inverse of an orthogonal matrix equals its transpose.

With the geometric perspective from the previous section, understanding the common properties of the matrix inverse becomes easy.

Non-Invertible Square Matrices

Let's start with the conclusion: a nonzero determinant is the necessary and sufficient condition for a matrix to be invertible.

A necessary and sufficient condition means: the condition holding guarantees that the conclusion holds (sufficiency), and the conclusion holding also, conversely, guarantees that the condition holds (necessity).

 What happens if a square matrix's determinant is 0?

Given the following 2×2 symmetric square matrix

$$\mathbf{P} = \begin{bmatrix} \frac{1}{4} & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & \frac{3}{4} \end{bmatrix}$$

Since whether a matrix is invertible depends on whether its determinant is 0, let's compute $\mathbf{P}$'s determinant:

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$$\det(\mathbf{P}) = \begin{vmatrix} \frac{1}{4} & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & \frac{3}{4} \end{vmatrix} = 0$$

Clearly, this 2×2 symmetric square matrix is not invertible.

To understand why this matrix is not invertible, we can analyze its effect from the perspective of projection.

As shown in Figure 1, the effect of the 2×2 symmetric square matrix $\mathbf{P}$ is to project a planar geometric shape onto a line through the origin; clearly, some information is lost after projection, so the original shape cannot be restored.

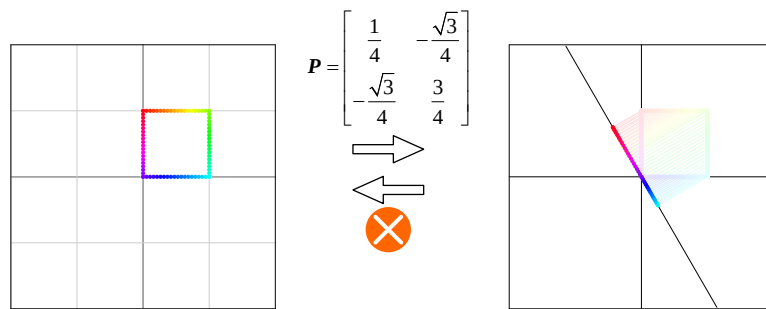


Figure 1. Projection in the Plane Is Not Invertible, 2×2 Matrix, First Example

In addition, please note that we can write the two column vectors of the projection matrix $\mathbf{P}$ as

$$\mathbf{p}_1 = \begin{bmatrix} \frac{1}{4} \\ -\frac{\sqrt{3}}{4} \end{bmatrix}, \quad \mathbf{p}_2 = (-\sqrt{3}) \times \begin{bmatrix} \frac{1}{4} \\ -\frac{\sqrt{3}}{4} \end{bmatrix} = -\sqrt{3} \mathbf{p}_1$$

That is, $\mathbf{p}_1$ and $\mathbf{p}_2$ are collinear and linearly dependent. This makes it easy to explain why $\mathbf{P}$'s determinant is 0 (the parallelogram spanned by $\mathbf{p}_1$ and $\mathbf{p}_2$ has area 0).

Please analyze on your own whether the determinant and column vectors of the non-invertible square matrix in Figure 2 are linearly dependent.

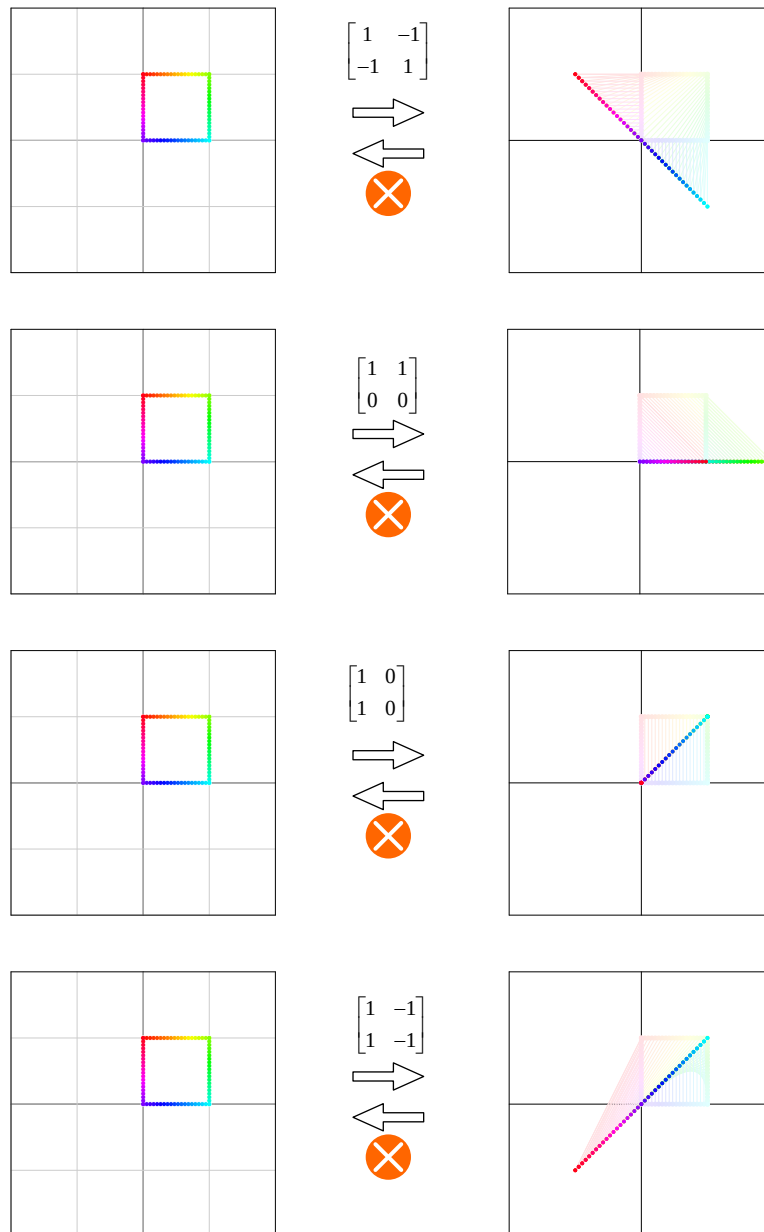


Figure 2. A Planar Geometric Transformation Is Not Invertible

Non-Square Matrices Are Not Invertible

If a matrix is not a square matrix, that is, its number of rows $\neq$ number of columns, then it is not invertible. Because the input and output dimensions of a non-square matrix's mapping differ, it cannot uniquely recover the original input.

Next, let's look at two examples.

As shown in Figure 3, the 3×2 matrix $\mathbf{A}$ maps two-dimensional space

$$\mathbf{Ax} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \\ a_{31}x_1 + a_{32}x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

This matrix maps all points in two-dimensional space to a certain two-dimensional plane within three-dimensional space, but it does not cover the entire three-dimensional space.

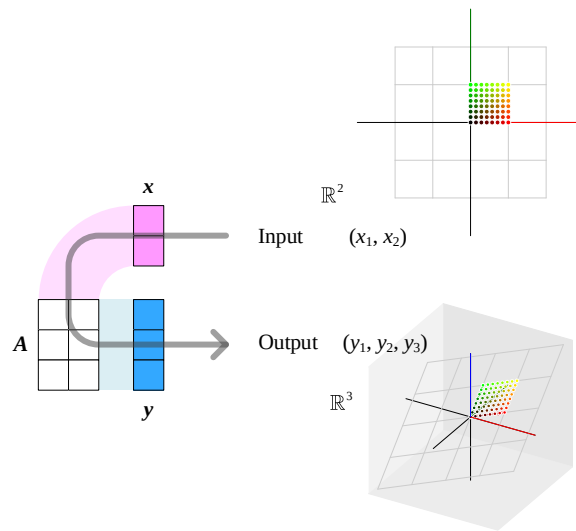


Figure 3. $Ax = y$, where the matrix A is 3×2

The information appears to have “increased” (from 2D to 3D), which means the original 2D coordinates cannot be uniquely recovered. The extra 1D dimension can vary freely, and multiple distinct 2D vectors may map to the same 3D vector. This is the geometric explanation for why a 3×2 matrix is not invertible.

Imagine drawing points on a sheet of paper (2D), then placing the paper into three-dimensional space. Although the points on the paper are all mapped into 3D space, they still lie on the same plane, and cannot be uniquely restored to 2D coordinates.

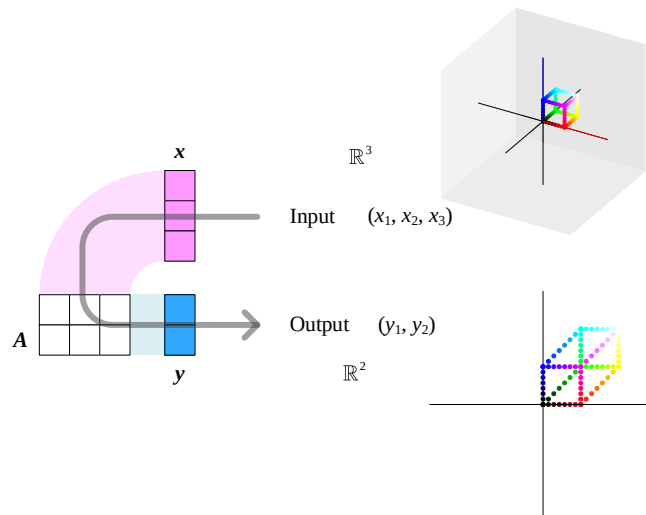
Let's look at another example. As shown in Figure 4, the 2×3 matrix A maps three-dimensional space to two-dimensional space.

$$Ax = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_{11}x_1 + b_{12}x_2 + b_{13}x_3 \\ b_{21}x_1 + b_{22}x_2 + b_{23}x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

The reason this is not invertible is that the mapping process causes information to be “lost,” going from 3D to 2D; different 3D vectors may project onto the same 2D vector, so the original 3D coordinates cannot be uniquely recovered.

Imagine you're in a room (3D) and you see an object's shadow on the wall (2D). Although you can see the shadow, you cannot uniquely determine the object's full three-dimensional shape from the shadow alone.

A non-square matrix is always non-invertible, because its input dimension $\neq$ output dimension, which causes information to be lost or gained, making it impossible for the inverse transformation to uniquely recover the original vector.

Figure 4. $Ax = y$, where the matrix A is 2×3

⚠ Note that non-invertible square matrices and non-square matrices also have their own kind of inverse – the pseudoinverse. The pseudoinverse is a generalized inverse for non-square or non-invertible matrices; it minimizes error and is used to solve least-squares problems. This is content the book will introduce later.

The Inverse of a Product of Two Matrices

If square matrices of the same shape A and B are both invertible, then

$$(AB)^{-1} = B^{-1}A^{-1}$$

From a geometric perspective, the matrix product $A @ B @ x$ is equivalent to: for a vector x , first apply the geometric operation of B , then apply the geometric operation corresponding to A .

Using the earlier example, R is the rotation matrix, and S is the scaling matrix

$$R = \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix}, \quad S = \begin{bmatrix} 3/2 & 0 \\ 0 & 2/3 \end{bmatrix}$$

As shown in Figure 5, $(R @ S) @ x$ represents the “one-step” composite geometric transformation.

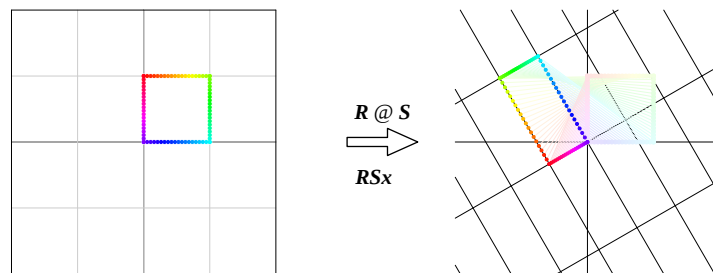


Figure 5. Composite Transformation

Clearly, as shown in Figure 6, $(R@S)$'s inverse operation is $(R@S)^{-1}$, that is,

$$(R@S)^{-1}@(R@S)@x = I@x = x$$

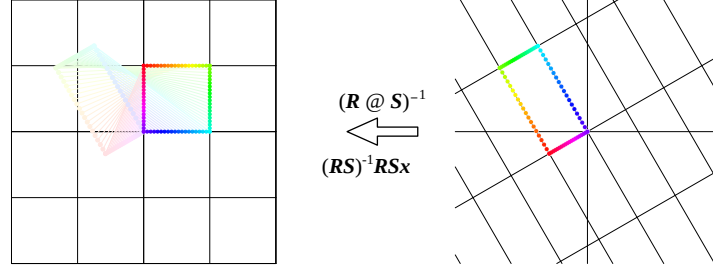


Figure 6. The Inverse of the Composite Transformation

Expanding the composite geometric operation $(R@S)$, the equation below represents “scale first, then rotate”

$$R@S@x = \underbrace{\begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix}}_{\text{Rotate}} @ \underbrace{\begin{bmatrix} 3/2 & 0 \\ 0 & 2/3 \end{bmatrix}}_{\text{Scale}} @ x$$

The geometric operation above corresponds to Figure 7.

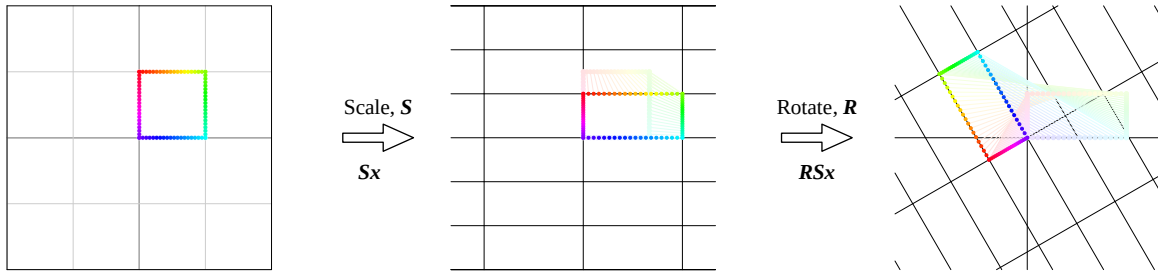


Figure 7. Scale First, Then Rotate

Looking at it in reverse, we need to first cancel out R , then cancel out S

$$S^{-1}@R^{-1}@R@S@x = I@x = x$$

The reverse sequence of operations corresponds to Figure 8.

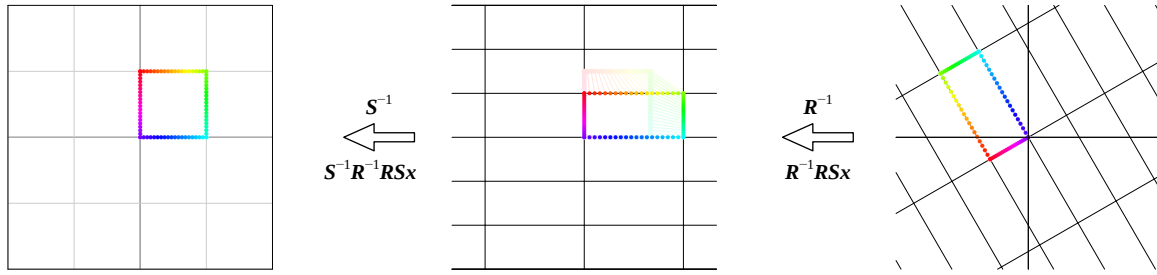


Figure 8. The Inverse Transformation: Reverse Rotation First, Then Reverse Scaling

What happens if we reverse-scale first, then reverse-rotate?

$$R @ S @ R^{-1} @ S^{-1} @ x$$

Figure 9 shows the inverse transformation done in the wrong order; clearly, we have not recovered the original shape (the unit square).

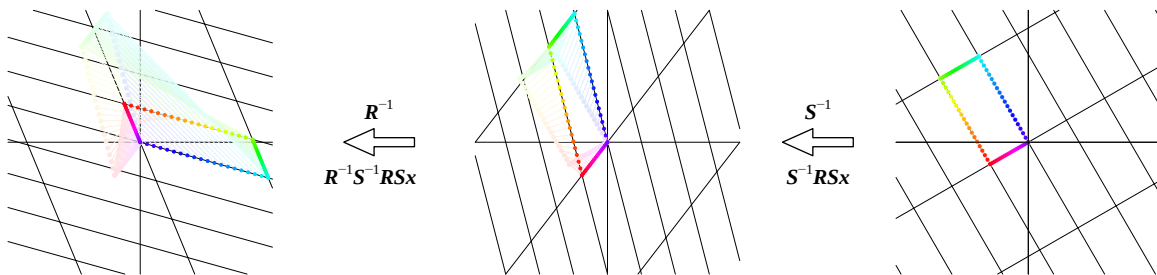


Figure 9. The Wrong Inverse Transformation: Reverse Scaling First, Then Reverse Rotation

The Inverse of a Diagonal Square Matrix

In the previous section, we already saw the effect of a diagonal matrix: each diagonal entry is an independent scaling factor.

For example, given the scaling matrix S

$$S = \begin{bmatrix} 2/3 & 0 \\ 0 & 3/2 \end{bmatrix}$$

As shown in Figure 10, S scales vectors in two-dimensional space along the x_1 axis and the x_2 axis, respectively.

S 's Inverse Matrix

$$S^{-1} = \begin{bmatrix} 3/2 & 0 \\ 0 & 2/3 \end{bmatrix}$$

is in fact simply the reciprocal of the diagonal entries.

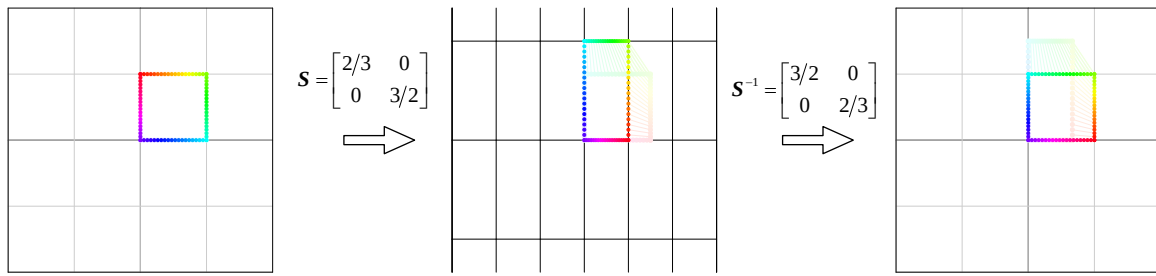


Figure 10. Scaling and the Inverse Scaling Operation in the Plane

The Inverse of a Symmetric Matrix

It's already clear that if $\mathbf{A}$ is a symmetric matrix, that is, it satisfies $\mathbf{A} = \mathbf{A}^T$, then its inverse matrix $\mathbf{A}^{-1}$ is also a symmetric matrix

$$(\mathbf{A}^{-1})^T = \mathbf{A}^{-1}$$

Let's give a simple proof

$$(\mathbf{A}^{-1})^T \mathbf{A} \mathbf{A}^{-1} = (\mathbf{A}^{-1})^T \mathbf{A}^T \mathbf{A}^{-1} = (\mathbf{A} \mathbf{A}^{-1})^T \mathbf{A}^{-1} = \mathbf{A}^{-1}$$

The Inverse of a Matrix Transpose

If $\mathbf{A}$ is invertible, then

$$(\mathbf{A}^{-1})^T = (\mathbf{A}^T)^{-1}$$

Let's give a simple proof

$$(\mathbf{A}^{-1})^T = (\mathbf{A}^{-1})^T \underbrace{\mathbf{A}^T (\mathbf{A}^T)^{-1}}_{\mathbf{I}} = ((\mathbf{A}^{-1})^T \mathbf{A}^T) (\mathbf{A}^T)^{-1} = (\mathbf{A} \mathbf{A}^{-1})^T (\mathbf{A}^T)^{-1} = (\mathbf{A}^T)^{-1}$$

To intuitively understand the geometric principle behind the equation, we need to draw on singular value decomposition (SVD).



Singular value decomposition is one of the important topics this book will explore later.

The Inverse of an Orthogonal Matrix

The previous section told us that if the matrix $\mathbf{A}$ is an orthogonal matrix, $\mathbf{A}$'s inverse is its transpose, that is,

$$\mathbf{A}^T = \mathbf{A}^{-1}$$

A rotation matrix is a type of orthogonal matrix; for example, the matrix $\mathbf{V}$ rotates a planar geometric shape counterclockwise about the origin by θ

$$\mathbf{V} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$\mathbf{V}$'s inverse matrix is:

$$\mathbf{V}^{-1} = \mathbf{V}^T = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$\mathbf{V}$ and $\mathbf{V}^T$'s product is the identity matrix

$$\mathbf{V} @ \mathbf{V}^T = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\mathbf{V}^T$ and $\mathbf{V}$'s product is also the identity matrix (with the same shape as the equation above)

$$\mathbf{V}^T @ \mathbf{V} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

It's worth noting that the identity matrix, mirror matrices, and permutation matrices are also all orthogonal matrices.

$k\mathbf{A}$'s Inverse

If the square matrix $\mathbf{A}$ is invertible, then $k\mathbf{A}$'s inverse is

$$(k\mathbf{A})^{-1} = \frac{1}{k} \mathbf{A}^{-1}$$

where k is nonzero. Simply put, when inverting the transformation, besides $\mathbf{A}^{-1}$, we also need to reverse the scaling – that is, multiply by $1/k$ – to cancel out the effect of k .

From another perspective, $k\mathbf{A}$ is equivalent to

$$k\mathbf{A} = k \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix} \mathbf{A} = \begin{bmatrix} k & & & \\ & k & & \\ & & \ddots & \\ & & & k \end{bmatrix} \mathbf{A}$$

Writing the $n \times n$ square matrix $\mathbf{A}$ as a set of row vectors, the equation above can be written as

$$\begin{bmatrix} k & & & \\ & k & & \\ & & \ddots & \\ & & & k \end{bmatrix} \mathbf{A} = \begin{bmatrix} k & & & \\ & k & & \\ & & \ddots & \\ & & & k \end{bmatrix} \begin{bmatrix} \mathbf{a}^{(1)} \\ \mathbf{a}^{(2)} \\ \vdots \\ \mathbf{a}^{(n)} \end{bmatrix} = \begin{bmatrix} k\mathbf{a}^{(1)} \\ k\mathbf{a}^{(2)} \\ \vdots \\ k\mathbf{a}^{(n)} \end{bmatrix}$$

We find that multiplying a diagonal square matrix by $\mathbf{A}$ scales $\mathbf{A}$'s row vectors by the diagonal square matrix's diagonal entries; as shown in Figure 11, if the diagonal square matrix's diagonal entries differ, then the scaling factor differs for each row.

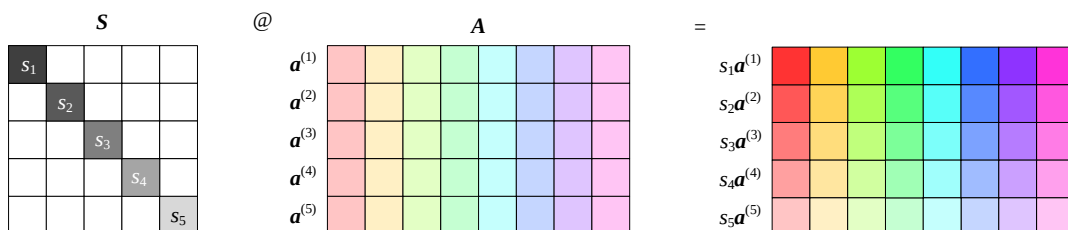


Figure 11. Scaling Each Row of the Wide Matrix $\mathbf{A}$

Conversely, if we write $k\mathbf{A}$ as

$$k\mathbf{A} = \mathbf{A} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix} \quad k\mathbf{A} = \mathbf{A} \begin{bmatrix} k & & & \\ & k & & \\ & & \ddots & \\ & & & k \end{bmatrix}$$

Now writing the $n \times n$ square matrix $\mathbf{A}$ as a set of column vectors, the equation above can be written as

$$\mathbf{A} \begin{bmatrix} k & & & \\ & k & & \\ & & \ddots & \\ & & & k \end{bmatrix} = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n] \begin{bmatrix} k & & & \\ & k & & \\ & & \ddots & \\ & & & k \end{bmatrix} = [k\mathbf{a}_1 \ k\mathbf{a}_2 \ \cdots \ k\mathbf{a}_n]$$

That is, as shown in Figure 12, $\mathbf{A}$ multiplied by the diagonal square matrix scales $\mathbf{A}$'s column vectors by the diagonal square matrix's diagonal entries.

Note that in Figure 11 and Figure 12, the matrix $\mathbf{A}$ has different shapes.

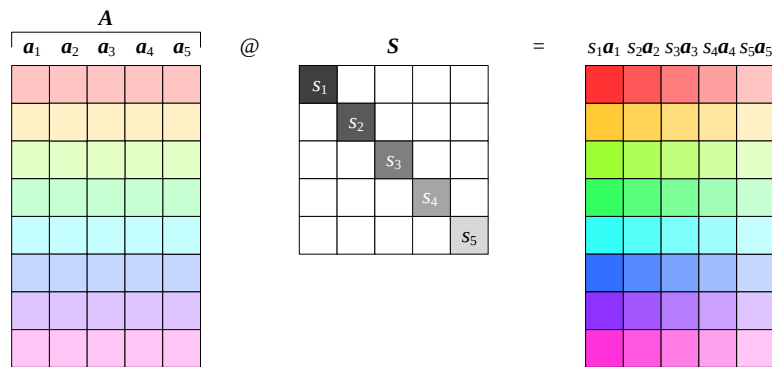


Figure 12. Scaling Each Column of the Tall Matrix $\mathbf{A}$

The Determinant of an Inverse Matrix

It's already clear that matrix $\mathbf{A}$'s determinant $\det(\mathbf{A})$ represents the transformation's scaling factor for area or volume. If matrix $\mathbf{A}$ is invertible, its inverse matrix's determinant is

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}$$

As shown in Figure 13, if $\mathbf{A}$ enlarges the area of the unit square by a factor of 2, then $\det(\mathbf{A}) = 2$.

Its inverse matrix $\mathbf{A}^{-1}$ shrinks the area to 1/2, so $\det(\mathbf{A}^{-1}) = 1/2$.

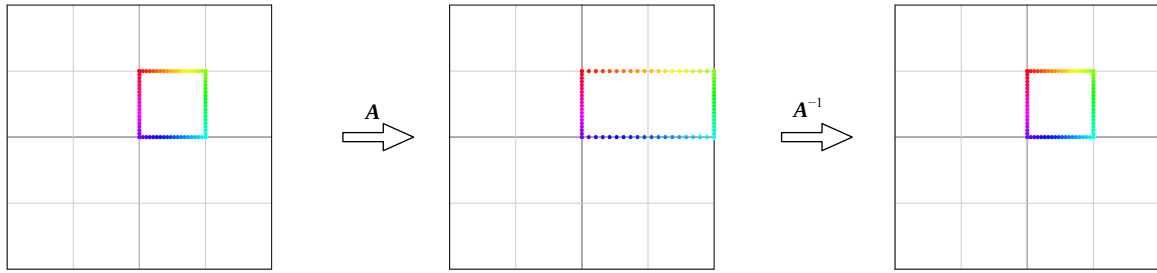


Figure 13. Scaling of Area



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Which of the following matrices are orthogonal matrices?

- ▶ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- ▶ $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$
- ▶ $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- ▶ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- ▶ $\begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}$
- ▶ $\begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$

Q2. Please compute the inverse of the following matrices, and explain their geometric meaning.

- ▶ $\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$
- ▶ $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- ▶ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- ▶ $\begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}$

Q3. Please indicate which square matrices are not invertible, and explain your reasoning.

 $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

 $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

 $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

 $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$